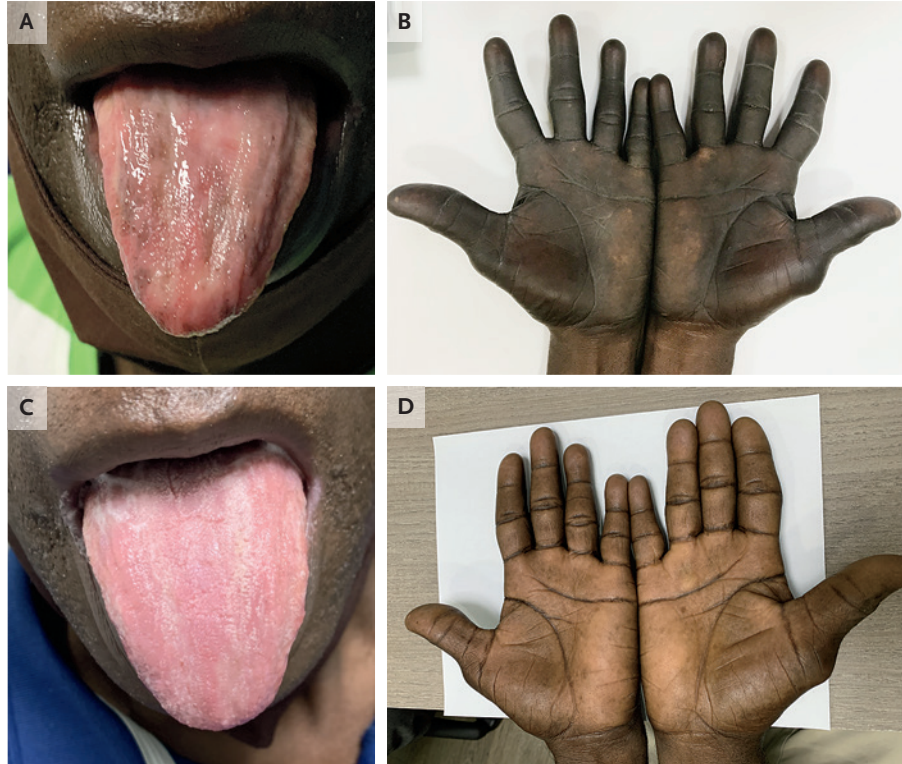


IMAGES IN CLINICAL MEDICINE

Stephanie V. Sherman, M.D., *Editor*Hyperpigmentation in Vitamin B₁₂ Deficiency

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A 59-YEAR-OLD MAN PRESENTED TO THE ENDOCRINOLOGY CLINIC WITH A 2-MONTH HISTORY OF FATIGUE and a 1-year history of progressively darkening skin on his palms and soles. He worked as a tile layer and had initially attributed his symptoms to occupational exposures. On physical examination, his tongue was smooth with patchy areas of mucosal darkening (Panel A). There was also hyperpigmentation on the palms (Panel B) and soles. Laboratory studies showed a hemoglobin level of 9.4 g per deciliter (reference range, 14 to 18), a mean corpuscular volume of 117 fl (reference range, 80 to 94), and mild leukopenia and thrombocytopenia. The vitamin B₁₂ level was 40 pg per milliliter (30 pmol per liter; reference range, 200 to 1100 pg per milliliter [150 to 810 pmol per liter]), and the serum level of intrinsic factor antibody was greater than 200 reference units per milliliter (reference value, <18), findings consistent with a diagnosis of vitamin B₁₂ deficiency due to autoimmune gastritis. Skin hyperpigmentation occurs in persons with vitamin B₁₂ deficiency owing to increased melanin synthesis and is seen more frequently in persons with darker skin. The hyperpigmentation typically reverses with treatment. The patient received treatment with parenteral vitamin B₁₂, and at follow-up 4 months after presentation, the hyperpigmentation had resolved (Panels C and D), as had his fatigue.

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